Mars Surface Power

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Moon to Mars Architecture Workshops

Introduction

Nuclear fission power selected as primary surface power generation technology for initial crewed missions to Mars

- 2024 Architecture Concept Review outcome
- Exercise in process as much as decision outcome
- Documented in Appendix B of the Architecture Definition Document

Technology Trade Space and Down Select

Nuclear Technologies

- **Fission surface power systems (FSP)**
- ~~Radioisotope thermoelectric generators (RTGs)~~

Non-Nuclear Technologies

- **Photovoltaic array with energy storage**
- ~~Primary batteries~~
- ~~Primary fuel cells~~
- ~~Wind power generation~~
- ~~Geothermal power generation~~
- ~~Biogeneration concepts~~

Summary of Decision

Nuclear fission power selected as primary surface power generation technology for initial crewed missions to Mars

Read the White Paper

https://bit.ly/3VN2Z1r